

Source Admission and Edition Scope

Edition: APS v26.7.30

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Status: normative candidate

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APS v26.7.30 begins by refusing a common failure in AI-assisted specification work: treating every available sentence as if it had equal authority. The repository contains a table of contents, generated HTML, AI-assistant doctrine, historical conversations, enterprise-architecture notes, an empty syntax file, and a prior complete manuscript. Those objects are evidence of different kinds. They cannot lawfully be merged until their provenance, purpose, and authority are classified.

The admitted corpus

The edition admits five source classes:

1. **Repository doctrine.** `CLAUDE.md`, the root README, module context files, and build instructions define the current repository fence. They control file placement, mutable versus immutable states, generated-output boundaries, and the mdBook build path.
2. **Current structural truth.** `specification-guide/src/SUMMARY.md` defines the inherited topic taxonomy. The generated search index and print edition demonstrate that the prior source chapters had headings but no body. The taxonomy is preserved; the emptiness is not.
3. **Historical design record.** `.service-colony.md`, `CUSTOM-GPT-CONTEXT.md`, planning conversations, and related archives preserve the evolution of APS from a Gherkin-like process notation toward a machine-readable operating constitution. They are informative evidence, not self-executing law.
4. **Prior complete manuscript.** `APS_2030_Complete_Manuscript.md` contributes the constitutional distinctions among intent, implementation, evidence, standing, falsifiers, receipts, and independent verification. Its repeated boilerplate is not imported as depth. Its propositions are normalized into this edition.
5. **External standards.** BCP 14 terminology, PROV-O, DCAT, SHACL, ODRL, BPMN, C4, OCEL 2.0, SLSA, and TOGAF are referenced only for the concerns they actually standardize. Reference does not imply equivalence or certification.

Admission function

Let the observed corpus be O . APS does not manufacture the edition directly from O ; it first constructs an admitted observation O^* .

```
O* = admit(O, authority, provenance, relevance, consistency, boundary)
A  =  $\mu(O^*)$ 
R  = receipt(A)
```

`admit` is not summarization. It is a typed partition:

```
admit(x)  $\in$  {NORMATIVE, INFORMATIVE, HISTORICAL, GENERATED, REFUSED, UNKNOWN}
```

A source is **NORMATIVE** only when it has explicit authority for the current coordinate. An **INFORMATIVE** source may explain a decision without authorizing it. A **HISTORICAL** source preserves lineage. A **GENERATED** source is a projection and must not be hand-edited as canonical input. A **REFUSED** source contradicts an admitted invariant or falls outside scope. **UNKNOWN** means authority or meaning has not been established; it is not permission to use the source silently.

What APS is

APS is a protocol for turning human and machine intent into bounded, reviewable, machine-readable work orders. It defines the declarations that must exist before downstream work receives standing: scope, owner, target state, admitted inputs, exclusions, acceptance checks, falsifiers, evidence requirements, authority, and expected receipts.

APS is not another sprint ceremony. It can be projected into Scrum, Kanban, SAFe, Jira, GitHub Issues, BPMN, Gantt, C4, PDDL, or other planning and representation systems. None of those projections exhausts the APS object. The same admitted work order may produce many views; no single view becomes the canonical law unless explicitly designated.

What APS is not

APS does not execute Gall checkpoints. It does not prove that implementation happened. It does not convert an author's confidence into standing. It does not grant an AI agent unrestricted authority. It does not treat a commit, test, review, or receipt as universal proof.

APS owns the **input model** and the **declaration boundary**. Gall owns checkpoint execution and the observation of whether the declared consequence became real. This distinction is constitutional and is formalized later in this book.

Normative language

The key words **MUST**, **MUST NOT**, **REQUIRED**, **SHALL**, **SHALL NOT**, **SHOULD**, **SHOULD NOT**, **RECOMMENDED**, **MAY**, and **OPTIONAL** are interpreted as BCP 14 terms only when written in uppercase. A requirement without an object, boundary, verifier, or failure consequence is not enforceable enough to receive normative standing.

Edition falsifier

The claim that v26.7.30 completes the APS book is falsified by any of the following at the release coordinate:

- a linked chapter is missing or heading-only;
- the APS/Gall ownership boundary is ambiguous;
- no machine-readable work-order schema exists;
- acceptance can be asserted without a falsifier or evidence coordinate;
- a failed or unsupported verifier is reported as success;
- generated mdBook output is treated as the source of truth;
- the book cannot be deterministically checked for structure and required invariants.

This edition therefore ships with schemas, a verifier, and a verifier receipt. The book is an admitted candidate only at the exact content digests recorded by that receipt.

APS Constitution

The constitution fixes the distinctions that every APS projection, schema, ticket, review, and implementation must preserve. Later chapters may specialize these laws but may not silently weaken them.

Root problem

Classical agile practice assumes that implementation capacity is scarce and coordination capacity is the principal bottleneck. Generative systems invert part of that premise. Candidate code, plans, documentation, and reviews can be produced faster than an organization can determine whether they correspond to the intended object. The new bottleneck is not generation. It is lawful admission.

APS addresses this by replacing narrative completion with protocol standing. Every material claim is represented as an object whose meaning is bounded by declared scope and whose promotion is controlled by evidence.

Constitutional objects

An APS system SHALL distinguish at least these object classes:

- **Intent** I : a desired consequence stated by an authorized source.
- **Observation** o : partial or stale information available to a producer.
- **Admitted observation** o^* : the subset of observation admitted under explicit law and boundary.
- **Work order** w : a deterministic contract describing a bounded transformation.
- **Candidate artifact** c : the proposed result of executing the work order.
- **Evidence bundle** E : coordinate-bound observations produced by verifiers.
- **Standing decision** s : a bounded classification of the candidate.
- **Receipt** R : a tamper-evident record binding operation, coordinate, evidence, decision, and lineage.
- **Repair packet** P : a typed delta that can produce a lawful successor work order.

These classes MUST NOT collapse into a single ticket description. In particular, $C \neq E$, $E \neq S$, and $S \neq R$.

Constitutional morphisms

```
parse      : Intent × Context → CandidateWorkOrder ∪ ParseRefusal
route      : CandidateWorkOrder → AuthorityDomain ∪ RouteRefusal
admit      : CandidateWorkOrder × Law → WorkOrder ∪ AdmissionRefusal
diagnose    : CandidateArtifact × WorkOrder → DeltaSet
verify     : CandidateArtifact × EvidenceContract → EvidenceBundle ∪
Unsupported
stand      : DeltaSet × EvidenceBundle × Policy → StandingDecision
receipt    : StandingDecision × Coordinate × Lineage → Receipt
repair     : Refusal × DeltaSet → SuccessorWorkOrder
```

A conforming implementation MUST make refusals first-class outputs. Exceptions and prose warnings are insufficient where the caller must distinguish malformed input, missing authority, unsupported verification, failed verification, and an implementation defect.

Six constitutional laws

Law 1 — Intent is not implementation

A requirement, plan, prompt, or ticket describes a target. It does not establish that a target exists. A work order may receive `Accepted` standing as a declaration while its implementation remains `UNKNOWN`.

Law 2 — Implementation is not evidence

Existence of code or documentation proves only existence at a coordinate. Evidence must identify the observer, command or method, environment, inputs, outputs, and scope of the claim.

Law 3 — Evidence is bounded

A passing unit test supports the tested behavior under its coordinate. It does not prove system-wide correctness, security, performance, or production suitability. Standing SHALL be no stronger than the weakest missing evidence class required by policy.

Law 4 — SELECT is not DO

Planning authority selects a lawful target and constructs a work order. An implementation agent performs the bounded delta. Selection does not actuate, and an implementer does not silently enlarge the selected scope.

Law 5 — Refusal preserves authority

A refused candidate **MUST NOT** mutate the accepted baseline. Refusal produces evidence and a repair path; it does not erase history or convert a failed candidate into a partial success.

Law 6 — Zero unreceipted actuation

Any operation that changes an authoritative or external state **SHALL** produce a receipt or an explicitly typed `UNSUPPORTED_RECEIPT` result. Logs alone are not receipts because they need not bind the declared object, authority, exact input, output digest, and decision.

Standing algebra

Operational standing uses the canonical vocabulary:

Standing	Meaning
ALIVE	The declared capability was observed executing at the stated coordinate and satisfied its bounded acceptance contract.
PARTIAL_ALIVE	A proper subset executed, or the evidence ladder closed only below the requested crown.
BLOCKED	A known external or prerequisite condition prevents lawful completion.
BUILD_BROKEN	The admitted tree does not compile, build, or assemble under the declared command.
UNKNOWN	Required observation has not been made or cannot be interpreted.
UNSUPPORTED	The available substrate cannot perform the requested operation or verification.

These states form neither a simple success/failure boolean nor a total order. `UNKNOWN` is not less successful than `BLOCKED`; it means a different epistemic condition. `UNSUPPORTED` is not refusal of the work; it is a capability boundary. `PARTIAL_ALIVE` is not permission to claim the crown.

Lifecycle state versus operational standing

Artifact lifecycle and operational standing are orthogonal:

```
Lifecycle ∈ {Draft, Provisional, InReview, Accepted, Final, Superseded,
Refused}
Standing ∈ {PARTIAL_ALIVE, ALIVE, BLOCKED, BUILD_BROKEN, UNKNOWN,
UNSUPPORTED}
```

An accepted specification can have `UNKNOWN` implementation standing. A draft prototype can be `ALIVE` at a narrow experimental coordinate. Conflating these axes creates documentation theater: the document's approval status becomes a substitute for observed execution.

Independence and authority

The producer of a material artifact SHALL NOT be the sole authority assigning its final standing. Independence is a graph property, not a job title. Two agents running the same prompt, model, toolchain, and hidden assumptions are not necessarily independent. An implementation SHOULD declare verifier diversity across at least one of: method, toolchain, model family, execution substrate, data source, or organizational authority.

Nonclaims

Conformance to APS does not prove moral legitimacy, regulatory approval, universal correctness, or freedom from defects. APS proves only that declared protocol objects were admitted, transformed, verified, classified, and receipted under a bounded coordinate.

Formal Semantics and Conformance Calculus

This chapter supplies the mathematical objects and decision functions needed to make APS falsifiable. The notation is intentionally bounded: it states exactly what a receipt or verifier establishes and prevents local evidence from being inflated into universal proof.

State model

Let an APS world be a labeled directed multigraph:

$$G = (V, E, \tau_V, \tau_E, \alpha)$$

V contains protocol objects. E contains typed relations such as `declares`, `supersedes`, `implements`, `verifies`, `refuses`, `authorizes`, `dependsOn`, and `receipts`. τ_V and τ_E assign object and relation types. α assigns attributes including content digests, versions, times, authorities, and coordinates.

A work order is not valid merely because its JSON parses. It is admitted when it satisfies structural, semantic, authority, and closure predicates:

$$\begin{aligned} \text{Admitted}(W) \Leftrightarrow & \text{Schema}(W) \wedge \text{Bounded}(W) \wedge \text{Authorized}(W) \\ & \wedge \text{Falsifiable}(W) \wedge \text{Verifiable}(W) \wedge \text{Closed}(W) \end{aligned}$$

$\text{Closed}(W)$ means every identifier required to interpret or execute the work order resolves within the admitted dependency boundary or is explicitly classified as an external prerequisite.

Observation closure

o is necessarily partial. APS therefore makes the admission boundary explicit:

$$O^* = \{x \in O \mid \text{admissible}(x, \text{policy}, \text{coordinate})\}$$

The manufacturing function μ is lawful only over O^* :

$$A = \mu(O^*)$$

The result A is not authoritative until a receipt binds the operation and its consequence:

$$R \vdash A = \mu(O^*)$$

The turnstile is intentionally bounded. R does not prove every possible property of A ; it proves that the recorded operation produced the recorded consequence from the recorded admitted input under the recorded coordinate.

Delta calculus

Given intended graph G_i and observed implementation graph G_o , APS computes a typed delta:

$$\Delta(G_i, G_o) = (M, X, C, U)$$

where:

- M is the set of missing required objects or edges;
- x is the set of extra forbidden objects or edges;
- c is the set of contradictory attributes or relations;
- u is the set of unevidenced claims.

The empty delta is necessary but not sufficient for **ALIVE**. Runtime evidence may still be absent. The standing function therefore considers both delta and evidence:

$$\text{stand}(\Delta, E, \Pi) \rightarrow \text{Standing}$$

Π is the policy profile. A documentation-only profile may accept structural evidence. A production-actuation profile may require unit, integration, end-to-end, chaos, stress, benchmark, and independent verifier evidence.

Falsifier semantics

Every normative claim q SHALL expose at least one falsifier $f(q)$ that operates on the same object and boundary:

$f(q) : \text{Observation} \rightarrow \{\text{supports}, \text{refutes}, \text{unknown}, \text{unsupported}\}$

Adjacency is not refutation. A failing test of a neighboring function does not refute q . A successful demonstration of a similar workflow does not support q . Same-object verification requires alignment of object identity, input domain, authority, closure, actuation path, and claimed consequence.

Evidence coordinates

Evidence is a tuple, not an adjective:

$\kappa = (\text{source}, \text{revision}, \text{tree}, \text{toolchain}, \text{target}, \text{features},$
 $\text{environment}, \text{command}, \text{inputDigest}, \text{outputDigest}, \text{time})$

Every evidence item SHALL bind to κ . Evidence without a source revision or input digest is vulnerable to drift. Evidence without a toolchain or target is not portable. Evidence without command and output is difficult to replay. Evidence without a claim identifier cannot be attributed to the object it supports.

Evidence ladder

APS defines an ordered verification program, not an automatic proof hierarchy:

$\text{unit} \rightarrow \text{integration} \rightarrow \text{e2e} \rightarrow \text{chaos} \rightarrow \text{stress} \rightarrow \text{benchmark} \rightarrow \text{verifier report}$

Each rung answers a different question. Unit tests isolate local laws. Integration tests establish composition. End-to-end tests exercise the selected path. Chaos tests challenge fault assumptions. Stress tests locate resource boundaries. Benchmarks characterize cost and latency. An independent verifier report audits the relationship among all prior evidence and the claim.

A policy MAY omit rungs for low-consequence artifacts, but the omission SHALL be explicit. A skipped rung is not a passed rung.

Transition calculus

Let L be lifecycle state and p a promotion request. Promotion is lawful only if:

```
promote(x, L1, L2) ⇔ allowed(L1, L2)
    ∧ criteria(x, L2)
    ∧ authority(p, L2)
    ∧ noBlockingRefusal(x)
```

Allowed transitions are:

```
Draft      → Provisional | Refused
Provisional → Draft | InReview | Refused
InReview   → Provisional | Accepted | Refused
Accepted   → Final | Superseded
Final      → Superseded
Refused    → Draft(successor only)
```

Direct Draft → Final promotion is refused. Reopening an accepted artifact edits a successor, not the accepted object. Content identity changes imply a new digest and therefore a new candidate coordinate.

Conformance profiles

- **APS-Core:** object identity, lifecycle, work-order schema, falsifier, typed outcome.
- **APS-Evidence:** Core plus evidence coordinates, independent verification, and receipts.
- **APS-Gall-Input:** Evidence plus deterministic checkpoint bindings and repair packet semantics.
- **APS-Actuation-Boundary:** Gall-Input plus broker authority, rollback, replay, and zero-unreceipted-actuation enforcement.

An implementation SHALL state the profile it claims. Claiming “APS compliant” without a profile is incomplete.

APS–Gall Boundary and Deterministic Work Orders

This boundary is the operational center of v26.7.30. APS manufactures admitted work orders; Gall executes and observes checkpoints. Preserving that separation prevents plans from impersonating consequences and implementation agents from granting themselves standing.

Preservation of the boundary

APS and Gall solve adjacent but non-equivalent problems.

APS : intent → admitted work order

Gall : admitted work order → checkpoint observation → standing → receipt

APS specifies what must be attempted and what evidence would count. Gall determines whether the attempt crossed a checkpoint. APS may name a Gall checkpoint, its dependencies, verifier command, expected receipt class, and falsifier. APS MUST NOT predeclare that the checkpoint is `ALIVE`.

SELECT and DO

The planner owns `SELECT` : choosing a bounded target, resolving dependencies, admitting exclusions, and constructing the work order. The coding or execution agent owns bounded `DO` : applying only the selected delta. Gall owns the checkpoint observation and standing assignment.

`SELECT(W) ≠ DO(W) ≠ VERIFY(W) ≠ PROMOTE(W)`

A single runtime may implement multiple roles, but role transitions SHALL be explicit and receipted. Hidden role collapse is self-certification.

Work-order normal form

A deterministic APS work order SHALL contain:

1. stable work-order and release identifiers;
2. source and exact base coordinate;
3. target consequence and bounded scope;
4. admitted observations and unresolved prerequisites;
5. required and forbidden deltas;
6. authority and role assignments;
7. acceptance commands and expected observations;
8. same-object falsifiers and negative fixtures;
9. evidence ladder and standing ceiling;
10. rollback, quarantine, or repair behavior;
11. Gall checkpoint identifiers and predecessor relations;
12. expected receipt and replay requirements;
13. explicit exclusions and nonclaims.

A ticket that says “implement feature X” is not in normal form. It lacks enough law to distinguish successful implementation, adjacent implementation, partial completion, unsupported verification, and scope drift.

Determinism criterion

A ticket is deterministic when two competent executors, starting from the same admitted coordinate, can identify the same permitted change set and the same acceptance boundary without negotiating hidden requirements.

This does not require a single implementation. APS permits reversible construction and multiple lawful candidates. It requires that candidate diversity remain inside declared constraints and that every candidate face the same falsifiers.

Gall checkpoint binding

```
[[gall.checkpoints]]
id = "APS-GALL-004"
predecessors = ["APS-GALL-003"]
claim = "work-order schema rejects unbounded scope"
standing_ceiling = "ALIVE"

[[gall.checkpoints.verification]]
kind = "unit"
command = "python3 specification-guide/scripts/verify_v26_7_30.py"
expected = "ALIVE"

[[gall.checkpoints.falsifiers]]
id = "unbounded-scope-accepted"
fixture = "fixtures/work_order_unbounded.json"
expected_refusal = "SCOPE_BOUNDARY_MISSING"
```

The checkpoint binding is a declaration. The resulting receipt belongs to Gall or the executing verifier. APS stores the expected shape and later links the observed receipt.

Repair packets

A failed checkpoint SHALL produce a repair packet that preserves the original work order and identifies the smallest typed delta:

```
work_order: aps-v26.7.30-book
checkpoint: APS-GALL-004
standing: BLOCKED
observed_delta: REQUIRED_FALSIFIER_MISSING
expected: every normative claim has a same-object falsifier
observed: claim aps.ticket.authority has no falsifier
required_repair: add a negative fixture and bind it to the claim
nonclaims:
  - no assertion that other checkpoints failed
```

Repair is a successor transformation. It does not rewrite the failed receipt or erase the candidate that produced it.

Jira and GitHub projections

APS work orders MAY project into Jira, GitHub Issues, Linear, or another tracker. The tracker object is a view. Canonical fields SHOULD be preserved in structured metadata, issue forms, front matter, or linked artifacts. Tracker status labels such as “Done” SHALL NOT override APS lifecycle or Gall standing.

Work-order falsifiers

A work order is refused when any of the following holds:

- the base coordinate cannot be resolved;
- scope contains open-ended verbs without a closure condition;
- acceptance references commands that do not exist and no unsupported path is declared;
- the same actor is producer and sole final verifier;
- a required Gall predecessor is absent;
- rollback or refusal behavior is missing for consequential actuation;
- exclusions are implicit;
- success can be claimed without a receipt.

Conformance, Verification, Receipts, and Replay

Conformance is a repeatable decision over named objects, not a stylistic judgment. This chapter defines how validators, evidence ladders, negative fixtures, receipts, and replay constrain what may be reported as ALIVE.

Conformance is executable

A specification that cannot distinguish conforming from nonconforming artifacts is explanatory literature, not an operational standard. APS therefore requires every claimed profile to expose a deterministic validator and a finite set of positive and negative fixtures.

Validator contract

A validator SHALL:

- identify the exact APS profile and schema versions;
- validate syntax before semantics;
- resolve referenced identifiers and dependencies;
- emit typed diagnostics with object paths;
- preserve all failures rather than stop at the first convenient error unless configured for fail-fast operation;
- distinguish UNKNOWN from UNSUPPORTED;
- produce a machine-readable report and a human-readable summary;
- hash the validated inputs and report content;
- exit nonzero for refused conformance.

Receipt model

A receipt binds a consequence cell:

```
 $\Gamma$  = (attempt, authority, admittedInput, operation, output,  
      evidence, standing, lineage, replayContract)
```

At minimum, a receipt contains:

```
{
  "schema": "aps.receipt.v26.7.30",
  "operation_id": "aps-book-verify",
  "subject": "sha256:...",
  "coordinate": {"revision": "...", "command": "..."},
  "standing": "ALIVE",
  "evidence": [{"kind": "structure", "digest": "sha256:..."}],
  "nonclaims": ["mdBook rendering not observed when mdbook is unavailable"]
}
```

A conventional log line cannot substitute because it usually lacks subject identity, authority, policy, standing, and replay obligations.

Replay

Replay answers whether the recorded operation still produces the recorded observation under the same or an explicitly translated coordinate.

$$\text{replay}(R, \kappa_2) \rightarrow \{\text{MATCH}, \text{DIVERGED}, \text{UNSUPPORTED}, \text{UNKNOWN}\}$$

A divergence does not automatically prove the original receipt fraudulent. The toolchain, source tree, dependency graph, clock, random seed, external service, or policy may have changed. The replay report SHALL identify the first divergent coordinate component it can observe.

Mutation and negative fixtures

Positive tests demonstrate that an admitted path can succeed. Negative fixtures demonstrate that forbidden paths are refused. Mutation adequacy challenges whether the verifier would detect a meaningful corruption.

For a claim set Q and mutant set M :

$$\text{mutation_score} = \text{killed}(M) / |M|$$

The score is interpretable only when mutants correspond to declared laws. Thousands of trivial mutants do not compensate for leaving the authority bypass, receipt omission, or scope widening mutant alive.

Book verifier profile

The v26.7.30 book validator checks:

- every SUMMARY link resolves;
- every chapter has substantive body and one top-level title;
- required constitutional distinctions occur in the admitted corpus;
- JSON schemas parse;
- duplicate boilerplate stays below a bounded threshold;
- no generated book/ output appears in the source manifest;
- a deterministic receipt records chapter count, word count, file digests, and standing.

Because `mdbook` is not guaranteed to be installed in every verifier environment, rendering is a separate capability. Lack of `mdbook` SHALL be reported as `UNSUPPORTED`, never silently counted as a successful build.

Independent review consequence

A verifier report SHALL name both the claims it supports and the questions it did not test. Review completion is not universal endorsement. Where reviewers disagree, APS preserves the competing observations, identifies the shared object coordinate, and routes the unresolved difference into a repair or authority decision rather than averaging incompatible conclusions.

Release gate

The release gate is conjunctive:

$$\text{ReleaseReady} \Leftrightarrow \text{StructureAlive} \wedge \text{SchemaAlive} \wedge \text{InvariantsAlive} \\ \wedge \text{ReceiptAlive} \wedge \text{noBlockingRefusal}$$

Rendering and link crawling MAY be additional required conjuncts in CI. A local release can be `PARTIAL_ALIVE` when structural verification succeeds but the rendering tool is unavailable. It becomes `ALIVE` for the full release profile only after the declared full verifier ladder closes.

Lifecycle and State Management

Lifecycle is the authority geometry of APS. Mutable states permit construction; immutable states preserve admitted baselines. Promotion is not a label change but a morphism with preconditions, authority, evidence, and a receipt. This part specifies the state machine, its refusal edges, and the invariant that a rejected candidate cannot corrupt the prior baseline.

Formal lens

$\text{transition}(a, s_1, s_2) = \text{authority} \wedge \text{criteria} \wedge \text{evidence} \wedge \text{receipt}$

Document States: Draft, Provisional, In Review, Accepted, Final

This section defines the canonical APS document-state algebra.

Draft

Requirement APS-05-01. The implementation SHALL represent **draft** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `DRAFT_MISSING`; an unavailable checking capability yields `UNSUPPORTED`; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict draft while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Provisional

Requirement APS-05-02. The implementation SHALL represent **provisional** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `PROVISIONAL_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict provisional while still receiving the standing or lifecycle promotion that the concern was intended to guard.

In Review

Requirement APS-05-03. The implementation SHALL represent **in review** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `IN_REVIEW_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict in review while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Accepted

Requirement APS-05-04. The implementation SHALL represent **accepted** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing

representation yields `ACCEPTED_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict accepted while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Final

Requirement APS-05-05. The implementation SHALL represent **final** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `FINAL_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict final while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Superseded and Refused

Requirement APS-05-06. The implementation SHALL represent **superseded and refused** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `SUPERSEDED_AND_REFUSED_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict superseded and refused while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Preserved constitutional material

The prior manuscript contained the following non-boilerplate material. v26.7.30 retains it as a specialized rule, example, definition, or algebra within the current coordinate.

State algebra

APS uses the following canonical states:

State	Meaning	May change content?	May authorize downstream work?
Draft	Unadmitted working material	Yes	No
Provisional	Coherent candidate with known gaps	Yes	Limited
In Review	Frozen candidate under challenge	Only through reviewed revision	No
Accepted	Admitted baseline for bounded use	Through supersession only	Yes
Final	Accepted and sealed for the declared release coordinate	No	Yes
Superseded	Former accepted or final artifact replaced by a successor	No	Historical only
Refused	Candidate rejected with typed reasons	No, but may spawn a new draft	No

The state machine SHALL reject silent transitions, including Draft directly to Final, and SHALL preserve the prior authoritative artifact when a candidate is refused.

Chapter-level minimum conformance

A conforming implementation of this section SHALL make every listed concern machine-readable, SHALL bind evidence to the exact subject coordinate, SHALL preserve the accepted baseline on refusal, and SHALL emit a receipt for successful promotion. The standing ceiling is **PARTIAL_ALIVE | ALIVE | BLOCKED | BUILD_BROKEN | UNKNOWN | UNSUPPORTED**; no chapter text authorizes a stronger claim than observed evidence.

Workflow for Document Lifecycle in AAP

Describes the end-to-end state transition workflow for APS artifacts.

Creation and admission

Requirement APS-05-07. The implementation SHALL represent **creation and admission** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `CREATION_AND_ADMISSION_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict creation and admission while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Independent challenge

Requirement APS-05-08. The implementation SHALL represent **independent challenge** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `INDEPENDENT_CHALLENGE_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict independent challenge while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Revision and reconciliation

Requirement APS-05-09. The implementation SHALL represent **revision and reconciliation** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `REVISION_AND_RECONCILIATION_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict revision and reconciliation while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Approval and sealing

Requirement APS-05-10. The implementation SHALL represent **approval and sealing** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `APPROVAL_AND_SEALING_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict approval and sealing while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Operational observation

Requirement APS-05-11. The implementation SHALL represent **operational observation** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `OPERATIONAL_OBSERVATION_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict operational observation while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Supersession

Requirement APS-05-12. The implementation SHALL represent **supersession** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `SUPERSESSION_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict supersession while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Chapter-level minimum conformance

A conforming implementation of this section SHALL make every listed concern machine-readable, SHALL bind evidence to the exact subject coordinate, SHALL preserve the accepted baseline on refusal, and SHALL emit a receipt for successful promotion. The standing ceiling is **PARTIAL_ALIVE | ALIVE | BLOCKED | BUILD_BROKEN | UNKNOWN | UNSUPPORTED**; no chapter text authorizes a stronger claim than observed evidence.

Transition Criteria Between Document States

Provides mechanically testable criteria for each lifecycle transition.

Required metadata

Requirement APS-05-13. The implementation SHALL represent **required metadata** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `REQUIRED_METADATA_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict required metadata while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Required evidence

Requirement APS-05-14. The implementation SHALL represent **required evidence** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `REQUIRED_EVIDENCE_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict required evidence while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Required reviewers

Requirement APS-05-15. The implementation SHALL represent **required reviewers** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `REQUIRED_REVIEWERS_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict required reviewers while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Refusal conditions

Requirement APS-05-16. The implementation SHALL represent **refusal conditions** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `REFUSAL_CONDITIONS_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict refusal conditions while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Automatic versus human-authorized transitions

Requirement APS-05-17. The implementation SHALL represent **automatic versus human-authorized transitions** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `AUTOMATIC_VERSUS_HUMAN_AUTHORIZED_TRANSITIONS_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict automatic versus human-authorized transitions while still receiving the standing or lifecycle

promotion that the concern was intended to guard.

Chapter-level minimum conformance

A conforming implementation of this section SHALL make every listed concern machine-readable, SHALL bind evidence to the exact subject coordinate, SHALL preserve the accepted baseline on refusal, and SHALL emit a receipt for successful promotion. The standing ceiling is **PARTIAL_ALIVE | ALIVE | BLOCKED | BUILD_BROKEN | UNKNOWN | UNSUPPORTED**; no chapter text authorizes a stronger claim than observed evidence.

Metadata, Naming, and Repository Structure

Metadata gives protocol objects stable identity outside conversation memory. Names and directories are projections for human and tool navigation, while embedded identifiers and content digests provide canonical identity. This part makes the repository tree legible without pretending path names alone carry semantic authority.

Formal lens

`identity(a) = {stableId, schema, version, digest}; path(a) is a projection`

Metadata-Rich File Naming Format for APS

This section defines deterministic file naming that carries enough context for humans and automation.

Canonical filename grammar

Requirement APS-06-01. The implementation SHALL represent **canonical filename grammar** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `CANONICAL_FILENAME_GRAMMAR_MISSING`; an unavailable checking capability yields `UNSUPPORTED`; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict canonical filename grammar while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Release and state tokens

Requirement APS-06-02. The implementation SHALL represent **release and state tokens** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `RELEASE_AND_STATE_TOKENS_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict release and state tokens while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Owner and artifact-class tokens

Requirement APS-06-03. The implementation SHALL represent **owner and artifact-class tokens** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `OWNER_AND_ARTIFACT_CLASS_TOKENS_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict owner and artifact-class tokens while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Digest suffixes

Requirement APS-06-04. The implementation SHALL represent **digest suffixes** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `DIGEST_SUFFIXES_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict digest suffixes while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Migration from legacy names

Requirement APS-06-05. The implementation SHALL represent **migration from legacy names** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `MIGRATION_FROM_LEGACY_NAMES_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict migration from legacy names while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Chapter-level minimum conformance

A conforming implementation of this section SHALL make every listed concern machine-readable, SHALL bind evidence to the exact subject coordinate, SHALL preserve the accepted baseline on refusal, and SHALL emit a receipt for successful promotion. The standing ceiling is **PARTIAL_ALIVE | ALIVE | BLOCKED | BUILD_BROKEN | UNKNOWN | UNSUPPORTED**; no chapter text authorizes a stronger claim than observed evidence.

Guidelines for Effective File Naming

Provides operational rules for names that remain stable across tools and repositories.

Human readability

Requirement APS-06-06. The implementation SHALL represent **human readability** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `HUMAN_READABILITY_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict human readability while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Machine parseability

Requirement APS-06-07. The implementation SHALL represent **machine parseability** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `MACHINE_PARSEABILITY_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict machine parseability while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Avoiding semantic duplication

Requirement APS-06-08. The implementation SHALL represent **avoiding semantic duplication** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `AVOIDING_SEMANTIC_DUPLICATION_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict avoiding semantic duplication while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Length limits and slugs

Requirement APS-06-09. The implementation SHALL represent **length limits and slugs** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `LENGTH_LIMITS_AND_SLUGS_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict length limits and slugs while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Collision handling

Requirement APS-06-10. The implementation SHALL represent **collision handling** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `COLLISION_HANDLING_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict collision handling while still receiving the standing or lifecycle promotion that the concern

was intended to guard.

Chapter-level minimum conformance

A conforming implementation of this section SHALL make every listed concern machine-readable, SHALL bind evidence to the exact subject coordinate, SHALL preserve the accepted baseline on refusal, and SHALL emit a receipt for successful promotion. The standing ceiling is **PARTIAL_ALIVE | ALIVE | BLOCKED | BUILD_BROKEN | UNKNOWN | UNSUPPORTED**; no chapter text authorizes a stronger claim than observed evidence.

Examples of Metadata-Rich File Names

Shows compliant and noncompliant names across common APS artifacts.

Product requirements

Requirement APS-06-11. The implementation SHALL represent **product requirements** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `PRODUCT_REQUIREMENTS_MISSING`; an unavailable checking capability yields `UNSUPPORTED`; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict product requirements while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Architecture decisions

Requirement APS-06-12. The implementation SHALL represent **architecture decisions** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `ARCHITECTURE_DECISIONS_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict architecture decisions while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Mutant definitions

Requirement APS-06-13. The implementation SHALL represent **mutant definitions** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `MUTANT_DEFINITIONS_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict mutant definitions while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Verification receipts

Requirement APS-06-14. The implementation SHALL represent **verification receipts** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `VERIFICATION_RECEIPTS_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict verification receipts while still receiving the standing or lifecycle promotion that the concern

was intended to guard.

Release ledgers

Requirement APS-06-15. The implementation SHALL represent **release ledgers** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `RELEASE_LEDGERS_MISSING`; an unavailable checking capability yields `UNSUPPORTED`; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict release ledgers while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Chapter-level minimum conformance

A conforming implementation of this section SHALL make every listed concern machine-readable, SHALL bind evidence to the exact subject coordinate, SHALL preserve the accepted baseline on refusal, and SHALL emit a receipt for successful promotion. The standing ceiling is **PARTIAL_ALIVE | ALIVE | BLOCKED | BUILD_BROKEN | UNKNOWN | UNSUPPORTED**; no chapter text authorizes a stronger claim than observed evidence.

Standard Directory Layout for APS

This section defines the normative repository layout for APS projects.

Constitution

Requirement APS-06-16. The implementation SHALL represent **constitution** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing

representation yields `CONSTITUTION_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict constitution while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Claims and falsifiers

Requirement APS-06-17. The implementation SHALL represent **claims and falsifiers** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `CLAIMS_AND_FALSIFIERS_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict claims and falsifiers while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Implementation manifests

Requirement APS-06-18. The implementation SHALL represent **implementation manifests** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `IMPLEMENTATION_MANIFESTS_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict implementation manifests while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Evidence and receipts

Requirement APS-06-19. The implementation SHALL represent **evidence and receipts** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `EVIDENCE_AND_RECEIPTS_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict evidence and receipts while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Generated artifacts

Requirement APS-06-20. The implementation SHALL represent **generated artifacts** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `GENERATED_ARTIFACTS_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict generated artifacts while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Archives and snapshots

Requirement APS-06-21. The implementation SHALL represent **archives and snapshots** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `ARCHIVES_AND_SNAPSHOTS_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict archives and snapshots while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Chapter-level minimum conformance

A conforming implementation of this section SHALL make every listed concern machine-readable, SHALL bind evidence to the exact subject coordinate, SHALL preserve the accepted baseline on refusal, and SHALL emit a receipt for successful promotion. The standing ceiling is **PARTIAL_ALIVE | ALIVE | BLOCKED | BUILD_BROKEN | UNKNOWN | UNSUPPORTED**; no chapter text authorizes a stronger claim than observed evidence.

Subdirectories by Document State and Agile Role

Explains when state and role should be represented through directories versus metadata.

Role-based ownership surfaces

Requirement APS-06-22. The implementation SHALL represent **role-based ownership surfaces** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `ROLE_BASED_OWNERSHIP_SURFACES_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict role-based ownership surfaces while still receiving the standing or lifecycle promotion that the concern was intended to guard.

State-based review queues

Requirement APS-06-23. The implementation SHALL represent **state-based review queues** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `STATE_BASED_REVIEW_QUEUES_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict state-based review queues while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Avoiding directory explosion

Requirement APS-06-24. The implementation SHALL represent **avoiding directory explosion** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `AVOIDING_DIRECTORY_EXPLOSION_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict avoiding directory explosion while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Cross-functional work products

Requirement APS-06-25. The implementation SHALL represent **cross-functional work products** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `CROSS_FUNCTIONAL_WORK_PRODUCTS_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict cross-functional work products while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Access control

Requirement APS-06-26. The implementation SHALL represent **access control** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `ACCESS_CONTROL_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict access control while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Chapter-level minimum conformance

A conforming implementation of this section SHALL make every listed concern machine-readable, SHALL bind evidence to the exact subject coordinate, SHALL preserve the accepted baseline on refusal, and SHALL emit a receipt for successful promotion. The standing ceiling is **PARTIAL_ALIVE | ALIVE | BLOCKED | BUILD_BROKEN | UNKNOWN | UNSUPPORTED**; no chapter text authorizes a stronger claim than observed evidence.

Managing Archived Versions and Snapshots

This section defines archival, snapshot, and historical replay requirements.

Immutable snapshots

Requirement APS-06-27. The implementation SHALL represent **immutable snapshots** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `IMMUTABLE_SNAPSHOTS_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict immutable snapshots while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Retention classes

Requirement APS-06-28. The implementation SHALL represent **retention classes** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `RETENTION_CLASSES_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict retention classes while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Snapshot manifests

Requirement APS-06-29. The implementation SHALL represent **snapshot manifests** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `SNAPSHOT_MANIFESTS_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict snapshot manifests while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Replayability

Requirement APS-06-30. The implementation SHALL represent **replayability** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `REPLAYABILITY_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict replayability while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Legal holds and deletion

Requirement APS-06-31. The implementation SHALL represent **legal holds and deletion** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `LEGAL_HOLDS_AND_DELETION_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict legal holds and deletion while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Chapter-level minimum conformance

A conforming implementation of this section SHALL make every listed concern machine-readable, SHALL bind evidence to the exact subject coordinate, SHALL preserve the accepted baseline on refusal, and SHALL emit a receipt for successful promotion. The standing ceiling is **PARTIAL_ALIVE | ALIVE | BLOCKED | BUILD_BROKEN | UNKNOWN | UNSUPPORTED**; no chapter text authorizes a stronger claim than observed evidence.

Versioning, Branch Worlds, and Reconciliation

A branch represents a candidate world, a commit identifies a coordinate, and a merge proposes reconciliation. None is proof of semantic preservation. APS versioning records both content evolution and authority evolution, then requires explicit handling of concurrent candidates and generated projections.

Formal lens

```
reconcile( $G_1, G_2, B$ )  $\rightarrow G_3 \cup \text{ConflictSet}$ ; commit  $\neq$  semantic proof
```

Overview of Version Control in APS

Explains how source control, artifact lineage, and standing interact.

Repository truth versus conversation memory

Requirement APS-07-01. The implementation SHALL represent **repository truth versus conversation memory** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `REPOSITORY_TRUTH_VERSUS_CONVERSATION_MEMORY_MISSING`; an unavailable checking capability yields `UNSUPPORTED`; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict repository truth versus conversation memory while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Branches as candidate worlds

Requirement APS-07-02. The implementation SHALL represent **branches as candidate worlds** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `BRANCHES_AS_CANDIDATE_WORLDS_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict branches as candidate worlds while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Commits as coordinates, not proof

Requirement APS-07-03. The implementation SHALL represent **commits as coordinates, not proof** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `COMMIT_AS_COORDINATES_NOT_PROOF_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict commits as coordinates, not proof while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Merge gates and reconciliation

Requirement APS-07-04. The implementation SHALL represent **merge gates and reconciliation** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `MERGE_GATES_AND_RECONCILIATION_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict merge gates and reconciliation while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Generated artifacts and source-of-truth boundaries

Requirement APS-07-05. The implementation SHALL represent **generated artifacts and source-of-truth boundaries** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `GENERATED_ARTIFACTS_AND_SOURCE_OF_TRUTH_BOUNDARIES_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict generated artifacts and source-of-truth boundaries while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Chapter-level minimum conformance

A conforming implementation of this section SHALL make every listed concern machine-readable, SHALL bind evidence to the exact subject coordinate, SHALL preserve the accepted baseline on refusal, and SHALL emit a receipt for successful promotion. The standing ceiling is **PARTIAL_ALIVE | ALIVE | BLOCKED | BUILD_BROKEN | UNKNOWN | UNSUPPORTED**; no chapter text authorizes a stronger claim than observed evidence.

Version Numbering System for APS

This section defines calendar, semantic, schema, and evidence versioning.

Release versions

Requirement APS-07-06. The implementation SHALL represent **release versions** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `RELEASE_VERSIONS_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict release versions while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Schema versions

Requirement APS-07-07. The implementation SHALL represent **schema versions** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `SCHEMA_VERSIONS_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict schema versions while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Artifact revisions

Requirement APS-07-08. The implementation SHALL represent **artifact revisions** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `ARTIFACT_REVISIONS_MISSING` ; an unavailable checking capability

yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict artifact revisions while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Standing coordinates

Requirement APS-07-09. The implementation SHALL represent **standing coordinates** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `STANDING_COORDINATES_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict standing coordinates while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Compatibility rules

Requirement APS-07-10. The implementation SHALL represent **compatibility rules** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `COMPATIBILITY_RULES_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict compatibility rules while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Chapter-level minimum conformance

A conforming implementation of this section SHALL make every listed concern machine-readable, SHALL bind evidence to the exact subject coordinate, SHALL preserve the accepted baseline on refusal, and SHALL emit a receipt for successful promotion. The standing ceiling is **PARTIAL_ALIVE | ALIVE | BLOCKED | BUILD_BROKEN | UNKNOWN | UNSUPPORTED**; no chapter text authorizes a stronger claim than observed evidence.

Handling and Reconciliation of Multiple Versions

Explains how APS handles concurrent proposals, forks, and conflicting baselines.

Candidate branches

Requirement APS-07-11. The implementation SHALL represent **candidate branches** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `CANDIDATE_BRANCHES_MISSING`; an unavailable checking capability yields `UNSUPPORTED`; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict candidate branches while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Three-way semantic reconciliation

Requirement APS-07-12. The implementation SHALL represent **three-way semantic reconciliation** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `THREE_WAY_SEMANTIC_RECONCILIATION_MISSING`; an unavailable

checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict three-way semantic reconciliation while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Authority over merges

Requirement APS-07-13. The implementation SHALL represent **authority over merges** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `AUTHORITY_OVER_MERGES_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict authority over merges while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Partial acceptance

Requirement APS-07-14. The implementation SHALL represent **partial acceptance** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `PARTIAL_ACCEPTANCE_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict partial acceptance while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Conflict receipts

Requirement APS-07-15. The implementation SHALL represent **conflict receipts** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `CONFLICT_RECEIPTS_MISSING`; an unavailable checking capability yields `UNSUPPORTED`; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict conflict receipts while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Chapter-level minimum conformance

A conforming implementation of this section SHALL make every listed concern machine-readable, SHALL bind evidence to the exact subject coordinate, SHALL preserve the accepted baseline on refusal, and SHALL emit a receipt for successful promotion. The standing ceiling is **PARTIAL_ALIVE | ALIVE | BLOCKED | BUILD_BROKEN | UNKNOWN | UNSUPPORTED**; no chapter text authorizes a stronger claim than observed evidence.

Version Control and Reconciliation Processes

Specifies the exact reconciliation pipeline for artifacts and implementation claims.

Intent model

Requirement APS-07-16. The implementation SHALL represent **intent model** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `INTENT_MODEL_MISSING`; an unavailable checking capability yields `UNSUPPORTED`; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict intent model while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Observed model

Requirement APS-07-17. The implementation SHALL represent **observed model** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `OBSERVED_MODEL_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict observed model while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Evidence model

Requirement APS-07-18. The implementation SHALL represent **evidence model** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `EVIDENCE_MODEL_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict evidence model while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Delta taxonomy

Requirement APS-07-19. The implementation SHALL represent **delta taxonomy** as an explicit protocol concern with a stable identifier, authority source, evidence expectation,

and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `DELTA_TAXONOMY_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict delta taxonomy while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Repair and reverification

Requirement APS-07-20. The implementation SHALL represent **repair and reverification** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `REPAIR_AND_REVERIFICATION_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict repair and reverification while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Preserved constitutional material

The prior manuscript contained the following non-boilerplate material. v26.7.30 retains it as a specialized rule, example, definition, or algebra within the current coordinate.

The reconciliation equation

APS defines three primary models:

- **I**, the intended model;
- **M**, the implemented model;
- **E**, the evidenced model.

It also permits an implementation report **R**, supplied by a human or agent.

The system computes:

$$\Delta IM = I \ominus M$$

$$\Delta ME = M \ominus E$$

$$\Delta RO = R \ominus O$$

where **O** is the independently observed repository or operational system. The operator $\ominus$ is a typed model difference, not a line diff. It identifies missing nodes, missing edges, forbidden nodes, forbidden edges, wrong properties, surviving legacy paths, missing evidence, and unreported collateral changes.

A reconciler SHALL never close a delta merely because a summary claims it was addressed. Closure requires the evidence declared by the governing claim.

Chapter-level minimum conformance

A conforming implementation of this section SHALL make every listed concern machine-readable, SHALL bind evidence to the exact subject coordinate, SHALL preserve the accepted baseline on refusal, and SHALL emit a receipt for successful promotion. The standing ceiling is **PARTIAL_ALIVE | ALIVE | BLOCKED | BUILD_BROKEN | UNKNOWN | UNSUPPORTED**; no chapter text authorizes a stronger claim than observed evidence.

Adversarial Review, Roles, and Approval

Adversarial review is cooperative falsification. It separates producer, challenger, verifier, integrator, and authority roles so that disagreement becomes structured evidence instead of interpersonal conflict. Approval is a bounded decision over a named object, not a general endorsement of the author or system.

Formal lens

```
standing = decide(claim, challenge, evidence, authority); producer ≠ sole  
verifier
```

Integrating Adversarial Agile Processes with GenAI

This section defines how constructive adversarial roles operate with generative systems.

Implementer, challenger, verifier, and integrator roles

Requirement APS-08-01. The implementation SHALL represent **implementer, challenger, verifier, and integrator roles** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `IMPLEMENTER_CHALLENGER_VERIFIER_AND_INTEGRATOR_ROLES_MISSING`; an unavailable checking capability yields `UNSUPPORTED`; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict implementer, challenger, verifier, and integrator roles while still receiving the standing or lifecycle promotion that the concern was intended to guard.

No self-certification

Requirement APS-08-02. The implementation SHALL represent **no self-certification** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `NO_SELF_CERTIFICATION_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict no self-certification while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Prompt claims as hypotheses

Requirement APS-08-03. The implementation SHALL represent **prompt claims as hypotheses** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `PROMPT_CLAIMS_AS_HYPOTHESES_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict prompt claims as hypotheses while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Repair packets instead of vague review requests

Requirement APS-08-04. The implementation SHALL represent **repair packets instead of vague review requests** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `REPAIR_PACKETS_INSTEAD_OF_VAGUE_REVIEW_REQUESTS_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict repair packets instead of vague review requests while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Escalation when models disagree

Requirement APS-08-05. The implementation SHALL represent **escalation when models disagree** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `ESCALATION_WHEN_MODELS_DISAGREE_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict escalation when models disagree while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Chapter-level minimum conformance

A conforming implementation of this section SHALL make every listed concern machine-readable, SHALL bind evidence to the exact subject coordinate, SHALL preserve the accepted baseline on refusal, and SHALL emit a receipt for successful promotion. The standing ceiling is **PARTIAL_ALIVE | ALIVE | BLOCKED | BUILD_BROKEN | UNKNOWN | UNSUPPORTED**; no chapter text authorizes a stronger claim than observed evidence.

Roles and Responsibilities in AAP

This section defines bounded roles and separation of duties.

Sponsor

Requirement APS-08-06. The implementation SHALL represent **sponsor** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `SPONSOR_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict sponsor while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Author or implementer

Requirement APS-08-07. The implementation SHALL represent **author or implementer** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `AUTHOR_OR_IMPLEMENTER_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict author or implementer while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Adversarial challenger

Requirement APS-08-08. The implementation SHALL represent **adversarial challenger** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing

representation yields `ADVERSARIAL_CHALLENGER_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict adversarial challenger while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Independent verifier

Requirement APS-08-09. The implementation SHALL represent **independent verifier** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `INDEPENDENT_VERIFIER_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict independent verifier while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Release integrator

Requirement APS-08-10. The implementation SHALL represent **release integrator** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `RELEASE_INTEGRATOR_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict release integrator while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Broker and auditor

Requirement APS-08-11. The implementation SHALL represent **broker and auditor** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `BROKER_AND_AUDITOR_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict broker and auditor while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Chapter-level minimum conformance

A conforming implementation of this section SHALL make every listed concern machine-readable, SHALL bind evidence to the exact subject coordinate, SHALL preserve the accepted baseline on refusal, and SHALL emit a receipt for successful promotion. The standing ceiling is **PARTIAL_ALIVE | ALIVE | BLOCKED | BUILD_BROKEN | UNKNOWN | UNSUPPORTED**; no chapter text authorizes a stronger claim than observed evidence.

Review Cycles, Feedback Loops, and Adversarial Testing

This section defines continuous review without endless conversational churn.

Review windows

Requirement APS-08-12. The implementation SHALL represent **review windows** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `REVIEW_WINDOWS_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict review windows while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Falsifier-driven review

Requirement APS-08-13. The implementation SHALL represent **falsifier-driven review** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `FALSIFIER_DRIVEN_REVIEW_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict falsifier-driven review while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Mutation and consequence testing

Requirement APS-08-14. The implementation SHALL represent **mutation and consequence testing** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `MUTATION_AND_CONSEQUENCE_TESTING_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict mutation and consequence testing while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Repair packets

Requirement APS-08-15. The implementation SHALL represent **repair packets** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `REPAIR_PACKETS_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict repair packets while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Stopping rules

Requirement APS-08-16. The implementation SHALL represent **stopping rules** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `STOPPING_RULES_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict stopping rules while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Chapter-level minimum conformance

A conforming implementation of this section SHALL make every listed concern machine-readable, SHALL bind evidence to the exact subject coordinate, SHALL preserve the accepted baseline on refusal, and SHALL emit a receipt for successful promotion. The standing ceiling is **PARTIAL_ALIVE | ALIVE | BLOCKED | BUILD_BROKEN | UNKNOWN | UNSUPPORTED**; no chapter text authorizes a stronger claim than observed evidence.

Approval Gates for Different Document States

This section defines approval requirements by artifact class and consequence level.

Low-risk documentation

Requirement APS-08-17. The implementation SHALL represent **low-risk documentation** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `LOW_RISK_DOCUMENTATION_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict low-risk documentation while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Architecture and schema changes

Requirement APS-08-18. The implementation SHALL represent **architecture and schema changes** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `ARCHITECTURE_AND_SCHEMA_CHANGES_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict architecture and schema changes while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Authority changes

Requirement APS-08-19. The implementation SHALL represent **authority changes** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `AUTHORITY_CHANGES_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict authority changes while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Production actuation

Requirement APS-08-20. The implementation SHALL represent **production actuation** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `PRODUCTION_ACTUATION_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict production actuation while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Emergency changes

Requirement APS-08-21. The implementation SHALL represent **emergency changes** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `EMERGENCY_CHANGES_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict emergency changes while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Chapter-level minimum conformance

A conforming implementation of this section SHALL make every listed concern machine-readable, SHALL bind evidence to the exact subject coordinate, SHALL preserve the accepted baseline on refusal, and SHALL emit a receipt for successful promotion. The standing ceiling is **PARTIAL_ALIVE | ALIVE | BLOCKED | BUILD_BROKEN | UNKNOWN | UNSUPPORTED**; no chapter text authorizes a stronger claim than observed evidence.

Governance, Compliance, and Auditability

Governance assigns who may define law, admit exceptions, authorize transitions, and revoke standing. Compliance maps external obligations into local claims and evidence without pretending APS itself grants certification. Auditability requires object-centric lineage capable of reconstructing why a state changed.

Formal lens

```
compliance(o) = map(obligation, controls, evidence, exceptions, coordinate)
```

Governance Model for APS Document Management

This section defines constitutional governance for APS repositories and programs.

Policy hierarchy

Requirement APS-09-01. The implementation SHALL represent **policy hierarchy** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `POLICY_HIERARCHY_MISSING`; an unavailable checking capability yields `UNSUPPORTED`; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict policy hierarchy while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Delegated authority

Requirement APS-09-02. The implementation SHALL represent **delegated authority** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `DELEGATED_AUTHORITY_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict delegated authority while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Exception admission

Requirement APS-09-03. The implementation SHALL represent **exception admission** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `EXCEPTION_ADMISSION_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict exception admission while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Governance as code

Requirement APS-09-04. The implementation SHALL represent **governance as code** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `GOVERNANCE_AS_CODE_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict governance as code while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Cross-organization federation

Requirement APS-09-05. The implementation SHALL represent **cross-organization federation** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `CROSS_ORGANIZATION_FEDERATION_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict cross-organization federation while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Chapter-level minimum conformance

A conforming implementation of this section SHALL make every listed concern machine-readable, SHALL bind evidence to the exact subject coordinate, SHALL preserve the accepted baseline on refusal, and SHALL emit a receipt for successful promotion. The standing ceiling is **PARTIAL_ALIVE | ALIVE | BLOCKED | BUILD_BROKEN | UNKNOWN | UNSUPPORTED**; no chapter text authorizes a stronger claim than observed evidence.

Compliance Requirements and Audit Trails

Maps APS evidence to compliance obligations without pretending equivalence to certification.

Control objectives

Requirement APS-09-06. The implementation SHALL represent **control objectives** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `CONTROL_OBJECTIVES_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict control objectives while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Evidence provenance

Requirement APS-09-07. The implementation SHALL represent **evidence provenance** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `EVIDENCE_PROVENANCE_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict evidence provenance while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Retention and access

Requirement APS-09-08. The implementation SHALL represent **retention and access** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `RETENTION_AND_ACCESS_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict retention and access while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Separation of duties

Requirement APS-09-09. The implementation SHALL represent **separation of duties** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `SEPARATION_OF_DUTIES_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict separation of duties while still receiving the standing or lifecycle promotion that the concern was intended to guard.

External assurance

Requirement APS-09-10. The implementation SHALL represent **external assurance** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `EXTERNAL_ASSURANCE_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict external assurance while still receiving the standing or lifecycle promotion that the concern

was intended to guard.

Chapter-level minimum conformance

A conforming implementation of this section SHALL make every listed concern machine-readable, SHALL bind evidence to the exact subject coordinate, SHALL preserve the accepted baseline on refusal, and SHALL emit a receipt for successful promotion. The standing ceiling is **PARTIAL_ALIVE | ALIVE | BLOCKED | BUILD_BROKEN | UNKNOWN | UNSUPPORTED**; no chapter text authorizes a stronger claim than observed evidence.

Maintaining Transparency with Audit Trails and Documentation

This section defines receipt DAGs, event evidence, and replayable audits.

Receipts versus logs

Requirement APS-09-11. The implementation SHALL represent **receipts versus logs** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `RECEIPTS_VERSUS_LOGS_MISSING`; an unavailable checking capability yields `UNSUPPORTED`; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict receipts versus logs while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Parent-child lineage

Requirement APS-09-12. The implementation SHALL represent **parent-child lineage** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `PARENT_CHILD_LINEAGE_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict parent-child lineage while still receiving the standing or lifecycle promotion that the concern was intended to guard.

OCEL-compatible events

Requirement APS-09-13. The implementation SHALL represent **ocel-compatible events** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `OCEL_COMPATIBLE_EVENTS_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict ocel-compatible events while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Tamper evidence

Requirement APS-09-14. The implementation SHALL represent **tamper evidence** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `TAMPER_EVIDENCE_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict tamper evidence while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Auditor experience

Requirement APS-09-15. The implementation SHALL represent **auditor experience** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `AUDITOR_EXPERIENCE_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict auditor experience while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Chapter-level minimum conformance

A conforming implementation of this section SHALL make every listed concern machine-readable, SHALL bind evidence to the exact subject coordinate, SHALL preserve the accepted baseline on refusal, and SHALL emit a receipt for successful promotion. The standing ceiling is **PARTIAL_ALIVE | ALIVE | BLOCKED | BUILD_BROKEN | UNKNOWN | UNSUPPORTED**; no chapter text authorizes a stronger claim than observed evidence.

Tools, Automation, and System Integration

Tools are replaceable implementations of protocol morphisms. APS specifies capabilities and interfaces rather than blessing a vendor stack. Automation may parse, validate, project, route, or request actuation, but it must preserve authority boundaries and typed failures.

Formal lens

`tool failure → UNSUPPORTED|BLOCKED, never truth-value collapse`

Recommended Tools for Managing APS Files

Provides a tool architecture rather than a frozen vendor list.

Source control

Requirement APS-10-01. The implementation SHALL represent **source control** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `SOURCE_CONTROL_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict source control while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Schema validation

Requirement APS-10-02. The implementation SHALL represent **schema validation** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `SCHEMA_VALIDATION_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict schema validation while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Artifact generation

Requirement APS-10-03. The implementation SHALL represent **artifact generation** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `ARTIFACT_GENERATION_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict artifact generation while still receiving the standing or lifecycle promotion that the concern was intended to guard.

AST and dependency scanning

Requirement APS-10-04. The implementation SHALL represent **ast and dependency scanning** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `AST_AND_DEPENDENCY_SCANNING_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict ast and dependency scanning while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Mutation testing

Requirement APS-10-05. The implementation SHALL represent **mutation testing** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `MUTATION_TESTING_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict mutation testing while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Receipt stores

Requirement APS-10-06. The implementation SHALL represent **receipt stores** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `RECEIPT_STORES_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict receipt stores while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Chapter-level minimum conformance

A conforming implementation of this section SHALL make every listed concern machine-readable, SHALL bind evidence to the exact subject coordinate, SHALL preserve the accepted baseline on refusal, and SHALL emit a receipt for successful promotion. The standing ceiling is **PARTIAL_ALIVE | ALIVE | BLOCKED | BUILD_BROKEN | UNKNOWN | UNSUPPORTED**; no chapter text authorizes a stronger claim than observed evidence.

Automation of Workflow Transitions and AI Integrations

This section defines safe automation of state transitions and agent workflows.

Trigger conditions

Requirement APS-10-07. The implementation SHALL represent **trigger conditions** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `TRIGGER_CONDITIONS_MISSING`; an unavailable checking capability yields `UNSUPPORTED`; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict trigger conditions while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Condition watches

Requirement APS-10-08. The implementation SHALL represent **condition watches** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `CONDITION_WATCHES_MISSING`; an unavailable checking capability

yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict condition watches while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Automated dispatch

Requirement APS-10-09. The implementation SHALL represent **automated dispatch** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `AUTOMATED_DISPATCH_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict automated dispatch while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Human authorization points

Requirement APS-10-10. The implementation SHALL represent **human authorization points** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `HUMAN_AUTHORIZATION_POINTS_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict human authorization points while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Failure isolation

Requirement APS-10-11. The implementation SHALL represent **failure isolation** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `FAILURE_ISOLATION_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict failure isolation while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Chapter-level minimum conformance

A conforming implementation of this section SHALL make every listed concern machine-readable, SHALL bind evidence to the exact subject coordinate, SHALL preserve the accepted baseline on refusal, and SHALL emit a receipt for successful promotion. The standing ceiling is **PARTIAL_ALIVE | ALIVE | BLOCKED | BUILD_BROKEN | UNKNOWN | UNSUPPORTED**; no chapter text authorizes a stronger claim than observed evidence.

Integration with Existing Agile and AI Systems

Shows how APS interoperates with Jira, GitHub, CI, LLM agents, and enterprise platforms.

Adapters and projections

Requirement APS-10-12. The implementation SHALL represent **adapters and projections** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `ADAPTERS_AND_PROJECTIONS_MISSING` ; an unavailable checking

capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict adapters and projections while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Bidirectional synchronization

Requirement APS-10-13. The implementation SHALL represent **bidirectional synchronization** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `BIDIRECTIONAL_SYNCHRONIZATION_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict bidirectional synchronization while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Avoiding dual authority

Requirement APS-10-14. The implementation SHALL represent **avoiding dual authority** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `AVOIDING_DUAL_AUTHORITY_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict avoiding dual authority while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Ticket generation

Requirement APS-10-15. The implementation SHALL represent **ticket generation** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `TICKET_GENERATION_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict ticket generation while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Evidence ingestion

Requirement APS-10-16. The implementation SHALL represent **evidence ingestion** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `EVIDENCE_INGESTION_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict evidence ingestion while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Chapter-level minimum conformance

A conforming implementation of this section SHALL make every listed concern machine-readable, SHALL bind evidence to the exact subject coordinate, SHALL preserve the accepted baseline on refusal, and SHALL emit a receipt for successful promotion. The standing ceiling is **PARTIAL_ALIVE | ALIVE | BLOCKED | BUILD_BROKEN | UNKNOWN | UNSUPPORTED**; no chapter text authorizes a stronger claim than observed evidence.

Operating Practice and Failure Discipline

Adoption begins with one closed claim–falsifier–evidence loop. Teams expand only after that loop is observable and replayable. This part catalogues the failure modes that make high-volume generation look productive while hiding missing standing.

Formal lens

`progress = closed evidence loops, not artifact count`

Guidelines for Creating and Managing APS Documents

Provides practical operating guidance for teams adopting APS.

Start with one claim loop

Requirement APS-11-01. The implementation SHALL represent **start with one claim loop** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `START_WITH_ONE_CLAIM_LOOP_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict start with one claim loop while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Keep claims bounded

Requirement APS-11-02. The implementation SHALL represent **keep claims bounded** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `KEEP_CLAIMS_BOUNDED_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict keep claims bounded while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Separate implementation and verification

Requirement APS-11-03. The implementation SHALL represent **separate implementation and verification** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `SEPARATE_IMPLEMENTATION_AND_VERIFICATION_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict separate implementation and verification while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Prefer typed refusals

Requirement APS-11-04. The implementation SHALL represent **prefer typed refusals** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `PREFER_TYPED_REFUSALS_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict prefer typed refusals while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Make nonclaims visible

Requirement APS-11-05. The implementation SHALL represent **make nonclaims visible** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `MAKE_NONCLAIMS_VISIBLE_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict make nonclaims visible while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Chapter-level minimum conformance

A conforming implementation of this section SHALL make every listed concern machine-readable, SHALL bind evidence to the exact subject coordinate, SHALL preserve the accepted baseline on refusal, and SHALL emit a receipt for successful promotion. The standing ceiling is **PARTIAL_ALIVE | ALIVE | BLOCKED | BUILD_BROKEN | UNKNOWN | UNSUPPORTED**; no chapter text authorizes a stronger claim than observed evidence.

Avoiding Common Pitfalls in APS Implementation

Catalogs recurring APS failure modes and their corrections.

Documentation theater

Requirement APS-11-06. The implementation SHALL represent **documentation theater** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `DOCUMENTATION_THEATER_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict documentation theater while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Overstated evidence

Requirement APS-11-07. The implementation SHALL represent **overstated evidence** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `OVERSTATED_EVIDENCE_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict overstated evidence while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Self-certification

Requirement APS-11-08. The implementation SHALL represent **self-certification** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `SELF_CERTIFICATION_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict self-certification while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Configuration drift

Requirement APS-11-09. The implementation SHALL represent **configuration drift** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `CONFIGURATION_DRIFT_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict configuration drift while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Generated-source editing

Requirement APS-11-10. The implementation SHALL represent **generated-source editing** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `GENERATED_SOURCE_EDITING_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict generated-source editing while still receiving the standing or lifecycle promotion that the

concern was intended to guard.

Ticket ambiguity

Requirement APS-11-11. The implementation SHALL represent **ticket ambiguity** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `TICKET_AMBIGUITY_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict ticket ambiguity while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Chapter-level minimum conformance

A conforming implementation of this section SHALL make every listed concern machine-readable, SHALL bind evidence to the exact subject coordinate, SHALL preserve the accepted baseline on refusal, and SHALL emit a receipt for successful promotion. The standing ceiling is **PARTIAL_ALIVE | ALIVE | BLOCKED | BUILD_BROKEN | UNKNOWN | UNSUPPORTED**; no chapter text authorizes a stronger claim than observed evidence.

Case Studies and Applied Analysis

The cases are analytical counterexamples, not testimonials. Each shows how the same APS laws behave under software, public-sector, privacy, and regulated-release constraints. The object is to expose where ordinary workflow labels would conceal authority or evidence gaps.

Formal lens

`case result = law applied to same-object evidence under domain boundary`

Case Studies, Lessons Learned, and Practical Examples

Demonstrates APS through bounded scenarios.

AI-generated Rust subsystem

Requirement APS-12-01. The implementation SHALL represent **ai-generated rust subsystem** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `AI_GENERATED_RUST_SUBSYSTEM_MISSING`; an unavailable checking capability yields `UNSUPPORTED`; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict ai-generated rust subsystem while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Municipal benefits workflow

Requirement APS-12-02. The implementation SHALL represent **municipal benefits workflow** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `MUNICIPAL_BENEFITS_WORKFLOW_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict municipal benefits workflow while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Privacy-preserving resource coalition

Requirement APS-12-03. The implementation SHALL represent **privacy-preserving resource coalition** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `PRIVACY_PRESERVING_RESOURCE_COALITION_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict privacy-preserving resource coalition while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Regulated release correction

Requirement APS-12-04. The implementation SHALL represent **regulated release correction** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `REGULATED_RELEASE_CORRECTION_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict regulated release correction while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Preserved constitutional material

The prior manuscript contained the following non-boilerplate material. v26.7.30 retains it as a specialized rule, example, definition, or algebra within the current coordinate.

Case study 1: AI-generated Rust authority subsystem

An AI agent reported that it had sealed a production certificate type. The repository contained the new type and passing tests, but AST reconciliation found a public `from_raw` constructor in a legacy module. APS classified the work as `PARTIALLY_IMPLEMENTED_BUT_REPORTED_COMPLETE` . A compile-fail client and a constructor-bypass mutant were added. The implementation earned standing only after the legacy constructor was removed and the designated oracle rejected external construction.

Case study 2: Municipal benefits workflow

A city described eligibility, appeal, privacy, and disbursement laws in an admitted model. The generated workflow separated recommendation from payment authority. During adversarial review, a consequence mutant removed the appeal notification while retaining the eligibility decision. The intended consequence oracle detected that the decision receipt existed without the required notice event. The release was blocked before deployment.

Case study 3: Privacy-preserving resource coalition

Several nonprofits coordinated transportation and appointment capacity without sharing unrestricted client records. Each participant emitted a bounded projection containing availability, geography, accessibility constraints, and a purpose-limited pseudonymous identifier. APS receipts bound the projection policy and detected a generated adapter that exposed a raw phone number field.

Case study 4: Regulated release correction

A production release passed its repository test suite but failed package reality verification because the verified policy file was excluded from the package. APS assigned `PACKAGE_MISMATCH`, prevented promotion, and produced an exact repair ticket identifying the missing package include rule.

Chapter-level minimum conformance

A conforming implementation of this section SHALL make every listed concern machine-readable, SHALL bind evidence to the exact subject coordinate, SHALL preserve the accepted baseline on refusal, and SHALL emit a receipt for successful promotion. The standing ceiling is **PARTIAL_ALIVE | ALIVE | BLOCKED | BUILD_BROKEN | UNKNOWN | UNSUPPORTED**; no chapter text authorizes a stronger claim than observed evidence.

Templates, Glossary, and Frequently Asked Questions

Templates are executable interfaces, not prose decorations. The glossary fixes distinctions that must survive tool projection. The FAQ fences APS against predictable category errors, especially the belief that a passing test, an accepted document, or an AI-generated artifact is automatically ALIVE.

Formal lens

`template validity = schema  $\wedge$  closure  $\wedge$  falsifier  $\wedge$  authority`

Glossary of Terms in APS

This section defines the normative vocabulary used throughout APS.

Admission

Requirement APS-13-01. The implementation SHALL represent **admission** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `ADMISSION_MISSING`; an unavailable checking capability yields `UNSUPPORTED`; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict admission while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Claim

Requirement APS-13-02. The implementation SHALL represent **claim** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `CLAIM_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict claim while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Falsifier

Requirement APS-13-03. The implementation SHALL represent **falsifier** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `FALSIFIER_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict falsifier while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Standing

Requirement APS-13-04. The implementation SHALL represent **standing** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing

representation yields `STANDING_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict standing while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Receipt

Requirement APS-13-05. The implementation SHALL represent **receipt** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `RECEIPT_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict receipt while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Broker

Requirement APS-13-06. The implementation SHALL represent **broker** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `BROKER_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict broker while still receiving the standing or lifecycle promotion that the concern was intended to guard.

ConstitutionIR

Requirement APS-13-07. The implementation SHALL represent **constitutionir** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `CONSTITUTIONIR_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict `constitutionir` while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Implementation delta

Requirement APS-13-08. The implementation SHALL represent **implementation delta** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `IMPLEMENTATION_DELTA_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict `implementation delta` while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Preserved constitutional material

The prior manuscript contained the following non-boilerplate material. v26.7.30 retains it as a specialized rule, example, definition, or algebra within the current coordinate.

Core terms

Admission — The act of accepting an observation, rule, artifact, or proposal into a bounded authoritative model.

Artifact contract — A versioned agreement between a producer and consumer that identifies schema, digests, ordering, required checks, and refusal behavior.

Broker — The only component authorized to perform a declared class of consequential action after validating an action envelope.

Claim — A bounded proposition about a system at a named coordinate.

Constitution — The machine-readable collection of laws, roles, falsifiers, evidence requirements, authority transitions, and gates governing a system.

ConstitutionIR — A canonical intermediate representation of the constitution suitable for compilation into checks, tests, mutants, gates, and receipts.

Evidence coordinate — The source, toolchain, configuration, environment, feature set, and artifact identity under which evidence applies.

Falsifier — A same-object observation that would demonstrate that a claim is false, incomplete, or unsupported.

Implementation delta — A typed difference between the intended model and observed implementation.

Receipt — A tamper-evident record binding an operation, inputs, outputs, tool coordinate, and result.

Standing — The bounded status assigned to a claim after evaluating its required evidence.

Chapter-level minimum conformance

A conforming implementation of this section SHALL make every listed concern machine-readable, SHALL bind evidence to the exact subject coordinate, SHALL preserve the accepted baseline on refusal, and SHALL emit a receipt for successful promotion. The standing ceiling is **PARTIAL_ALIVE | ALIVE | BLOCKED | BUILD_BROKEN | UNKNOWN | UNSUPPORTED**; no chapter text authorizes a stronger claim than observed evidence.

Templates and Examples for APS Artifacts

Provides reusable templates for claims, implementation manifests, reviews, and receipts.

Claim template

Requirement APS-13-09. The implementation SHALL represent **claim template** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `CLAIM_TEMPLATE_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict claim template while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Implementation claim template

Requirement APS-13-10. The implementation SHALL represent **implementation claim template** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `IMPLEMENTATION_CLAIM_TEMPLATE_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict implementation claim template while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Adversarial review template

Requirement APS-13-11. The implementation SHALL represent **adversarial review template** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `ADVERSARIAL_REVIEW_TEMPLATE_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict adversarial review template while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Receipt template

Requirement APS-13-12. The implementation SHALL represent **receipt template** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `RECEIPT_TEMPLATE_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict receipt template while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Release gate template

Requirement APS-13-13. The implementation SHALL represent **release gate template** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing

representation yields `RELEASE_GATE_TEMPLATE_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict release gate template while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Preserved constitutional material

The prior manuscript contained the following non-boilerplate material. v26.7.30 retains it as a specialized rule, example, definition, or algebra within the current coordinate.

Claim template

```
[[claims]]
```

```
id = "payments.refund.requires-approval"  
owner = "payments-implementation"  
verifier = "payments-independent-verifier"  
gates = ["release.G4"]
```

```
[claims.scope]
```

```
package = "payments-core"  
symbol = "RefundService::execute"
```

```
[claims.law]
```

```
kind = "authority-transition"  
required_predecessor = "ApprovedRefund"  
forbidden_input = "RefundRequest"
```

```
[claims.evidence]
```

```
required = ["ast", "compile-test", "runtime-test", "mutation", "receipt"]
```

Implementation claim template

```
schema = "aps-implementation-claim.v1"
mission = "refund-approval-2029-184"
implemented_claims = ["payments.refund.requires-approval"]
changed_files = ["crates/payments-core/src/refund.rs"]
changed_symbols = ["RefundService::execute"]
tests_added = ["refund_without_approval_is_refused"]
nonclaims = ["universal fraud prevention"]
```

Repair packet template

```
claim: payments.refund.requires-approval
standing: BLOCKED
observed_delta: FORBIDDEN_EDGE
location: crates/payments-core/src/refund.rs:142
expected: RefundService::execute accepts ApprovedRefund only
observed: RefundService::execute accepts RefundRequest
owner: payments-implementation
verifier: payments-independent-verifier
required_repair: remove the raw request path and preserve the refusal tests
```

Chapter-level minimum conformance

A conforming implementation of this section SHALL make every listed concern machine-readable, SHALL bind evidence to the exact subject coordinate, SHALL preserve the accepted baseline on refusal, and SHALL emit a receipt for successful promotion. The standing ceiling is **PARTIAL_ALIVE | ALIVE | BLOCKED | BUILD_BROKEN | UNKNOWN | UNSUPPORTED**; no chapter text authorizes a stronger claim than observed evidence.

Frequently Asked Questions about APS

Answers common adoption, governance, AI, and tooling questions.

Is APS another agile framework

Requirement APS-13-14. The implementation SHALL represent **is aps another agile framework** as an explicit protocol concern with a stable identifier, authority source,

evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `IS_APS_ANOTHER_AGILE_FRAMEWORK_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict is aps another agile framework while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Does APS require AI

Requirement APS-13-15. The implementation SHALL represent **does aps require ai** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `DOES_APS_REQUIRE_AI_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict does aps require ai while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Does a passing test mean ALIVE

Requirement APS-13-16. The implementation SHALL represent **does a passing test mean alive** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `DOES_A_PASSING_TEST_MEAN_ALIVE_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict does a passing test mean alive while still receiving the standing or lifecycle promotion that the concern was intended to guard.

How much formal proof is required

Requirement APS-13-17. The implementation SHALL represent **how much formal proof is required** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `HOW_MUCH_FORMAL_PROOF_IS_REQUIRED_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict how much formal proof is required while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Can humans override the system

Requirement APS-13-18. The implementation SHALL represent **can humans override the system** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `CAN_HUMANS_OVERRIDE_THE_SYSTEM_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict can humans override the system while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Chapter-level minimum conformance

A conforming implementation of this section SHALL make every listed concern machine-readable, SHALL bind evidence to the exact subject coordinate, SHALL preserve the accepted baseline on refusal, and SHALL emit a receipt for successful promotion. The standing ceiling is **PARTIAL_ALIVE | ALIVE | BLOCKED | BUILD_BROKEN | UNKNOWN | UNSUPPORTED**; no chapter text authorizes a stronger claim than observed evidence.

Change Log and Standards Map

This part records the v26.7.30 semantic break from the hollow repository scaffold and maps APS to external standards by concern. The mappings are deliberately non-equivalent: PROV-O can express provenance, SHACL can validate graph constraints, and BPMN can depict process flow, but none individually supplies the APS standing calculus.

Formal lens

`mapping(APS, standard)` is concern-preserving, not equivalence-preserving

Record of Updates and Changes to the APS

This section defines and initializes the standard change log.

Change classes

Requirement APS-14-01. The implementation SHALL represent **change classes** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `CHANGE_CLASSES_MISSING`; an unavailable checking capability yields `UNSUPPORTED`; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict change classes while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Approval requirements

Requirement APS-14-02. The implementation SHALL represent **approval requirements** as an explicit protocol concern with a stable identifier, authority source, evidence expectation,

and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `APPROVAL_REQUIREMENTS_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict approval requirements while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Migration notes

Requirement APS-14-03. The implementation SHALL represent **migration notes** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `MIGRATION_NOTES_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict migration notes while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Prior 30.1.1 manuscript baseline

Requirement APS-14-04. The implementation SHALL represent **prior 30.1.1 manuscript baseline** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `PRIOR_30_1_1_MANUSCRIPT_BASELINE_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict prior 30.1.1 manuscript baseline while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Chapter-level minimum conformance

A conforming implementation of this section SHALL make every listed concern machine-readable, SHALL bind evidence to the exact subject coordinate, SHALL preserve the accepted baseline on refusal, and SHALL emit a receipt for successful promotion. The standing ceiling is **PARTIAL_ALIVE | ALIVE | BLOCKED | BUILD_BROKEN | UNKNOWN | UNSUPPORTED**; no chapter text authorizes a stronger claim than observed evidence.

Related Standards, Frameworks, and Methodologies

Positions APS relative to established standards and methods.

Scrum and Kanban

Requirement APS-14-05. The implementation SHALL represent **scrum and kanban** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `SCRUM_AND_KANBAN_MISSING`; an unavailable checking capability yields `UNSUPPORTED`; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict scrum and kanban while still receiving the standing or lifecycle promotion that the concern was intended to guard.

CMMI and ISO management systems

Requirement APS-14-06. The implementation SHALL represent **cmmi and iso management systems** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `CMMI_AND_ISO_MANAGEMENT_SYSTEMS_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict `cmmi` and `iso` management systems while still receiving the standing or lifecycle promotion that the concern was intended to guard.

TOGAF and architecture governance

Requirement APS-14-07. The implementation SHALL represent **togaf and architecture governance** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `TOGAF_AND_ARCHITECTURE_GOVERNANCE_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict `togaf` and `architecture governance` while still receiving the standing or lifecycle promotion that the concern was intended to guard.

PROV-O and OCEL

Requirement APS-14-08. The implementation SHALL represent **prov-o and ocel** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `PROV_O_AND_OCEL_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict `prov-o` and `ocel` while still receiving the standing or lifecycle promotion that the concern was intended to guard.

SLSA and software supply chain

Requirement APS-14-09. The implementation SHALL represent **slsa and software supply chain** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `SLSA_AND_SOFTWARE_SUPPLY_CHAIN_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict slsa and software supply chain while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Chapter-level minimum conformance

A conforming implementation of this section SHALL make every listed concern machine-readable, SHALL bind evidence to the exact subject coordinate, SHALL preserve the accepted baseline on refusal, and SHALL emit a receipt for successful promotion. The standing ceiling is **PARTIAL_ALIVE | ALIVE | BLOCKED | BUILD_BROKEN | UNKNOWN | UNSUPPORTED**; no chapter text authorizes a stronger claim than observed evidence.

Recommended Reading on Agile and AI Integration

Provides a curated conceptual reading path without making APS dependent on any one source.

Agile and lean foundations

Requirement APS-14-10. The implementation SHALL represent **agile and lean foundations** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `AGILE_AND_LEAN_FOUNDATIONS_MISSING` ; an unavailable checking

capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict agile and lean foundations while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Formal methods and contracts

Requirement APS-14-11. The implementation SHALL represent **formal methods and contracts** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `FORMAL_METHODS_AND_CONTRACTS_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict formal methods and contracts while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Distributed systems and authority

Requirement APS-14-12. The implementation SHALL represent **distributed systems and authority** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `DISTRIBUTED_SYSTEMS_AND_AUTHORITY_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict distributed systems and authority while still receiving the standing or lifecycle promotion that the concern was intended to guard.

AI assurance and evaluation

Requirement APS-14-13. The implementation SHALL represent **ai assurance and evaluation** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `AI_ASSURANCE_AND_EVALUATION_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict ai assurance and evaluation while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Pattern languages

Requirement APS-14-14. The implementation SHALL represent **pattern languages** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `PATTERN_LANGUAGES_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict pattern languages while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Chapter-level minimum conformance

A conforming implementation of this section SHALL make every listed concern machine-readable, SHALL bind evidence to the exact subject coordinate, SHALL preserve the accepted baseline on refusal, and SHALL emit a receipt for successful promotion. The standing ceiling is **PARTIAL_ALIVE | ALIVE | BLOCKED | BUILD_BROKEN | UNKNOWN | UNSUPPORTED**; no chapter text authorizes a stronger claim than observed evidence.

External Resources and Further Learning

This section defines categories of external resources and a process for maintaining them.

Official standards

Requirement APS-14-15. The implementation SHALL represent **official standards** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `OFFICIAL_STANDARDS_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict official standards while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Primary technical documentation

Requirement APS-14-16. The implementation SHALL represent **primary technical documentation** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `PRIMARY_TECHNICAL_DOCUMENTATION_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict primary technical documentation while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Research repositories

Requirement APS-14-17. The implementation SHALL represent **research repositories** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `RESEARCH_REPOSITORIES_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict research repositories while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Tool ecosystems

Requirement APS-14-18. The implementation SHALL represent **tool ecosystems** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing representation yields `TOOL_ECOSYSTEMS_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict tool ecosystems while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Community practice

Requirement APS-14-19. The implementation SHALL represent **community practice** as an explicit protocol concern with a stable identifier, authority source, evidence expectation, and failure consequence. It SHALL NOT leave the concern solely in narrative convention or infer it from a tracker status.

Operational test. A verifier MUST be able to locate the representation, resolve its dependencies, and produce one of the canonical standing outcomes. Missing

representation yields `COMMUNITY_PRACTICE_MISSING` ; an unavailable checking capability yields `UNSUPPORTED` ; contradictory evidence yields a typed refusal rather than a softened success.

Falsifier. Conformance is refuted when a same-coordinate candidate can omit or contradict community practice while still receiving the standing or lifecycle promotion that the concern was intended to guard.

Chapter-level minimum conformance

A conforming implementation of this section SHALL make every listed concern machine-readable, SHALL bind evidence to the exact subject coordinate, SHALL preserve the accepted baseline on refusal, and SHALL emit a receipt for successful promotion. The standing ceiling is **PARTIAL_ALIVE | ALIVE | BLOCKED | BUILD_BROKEN | UNKNOWN | UNSUPPORTED**; no chapter text authorizes a stronger claim than observed evidence.